Chapter 7 Estimating Parameters and Determining Sample Sizes
Estimating a Population Mean (7.2)

Objectives

In this lesson, we will learn how to

- use a sample mean as a point estimate of a population mean;
- construct a confidence interval for a population mean when σ is not known;
- use the Student t distribution and degrees of freedom;
- choose between the t distribution and the normal z distribution;
- find the sample size needed to estimate a population mean.

1 Main Goal

In this section, our main goal is to use a sample mean to make an inference about the corresponding population mean.

Three Main Ideas

- Point estimate: The sample mean $\bar{x}$ is the best single-value estimate of the population mean μ.
- Confidence interval: We use sample data to estimate a range of likely values for μ.
- Sample size: We find how large a sample should be to estimate μ with a desired accuracy.

2 Notation

Notation for Estimating a Population Mean

μ = population mean
$\bar{x}$ = sample mean
s = sample standard deviation
σ = population standard deviation
n = number of sample values
E = margin of error

3 Confidence Interval for μ When σ Is Not Known

In most real applications, the population standard deviation σ is not known. In that case, we use the sample standard deviation s and the Student t distribution.

Requirements

To construct a confidence interval for μ when σ is not known:

1. The sample is a simple random sample.
2. Either the population is normally distributed, or $n > 30$.

Confidence Interval for μ

When σ is not known,

$$\bar{x} - E < \mu < \bar{x} + E,$$

where

$$E = t_{\alpha/2}\frac{s}{\sqrt{n}}.$$

Use

$$\text{df} = n - 1.$$

4 Confidence Level and Critical Value

A confidence interval is associated with a confidence level, such as 95%.

For a 95% confidence level,

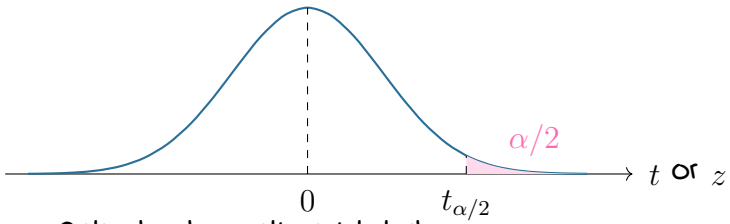
$$\alpha = 1 - 0.95 = 0.05.$$

Because the confidence interval has two tails,

$$\frac{\alpha}{2} = 0.025.$$

Critical Value $t_{\alpha/2}$

The value $t_{\alpha/2}$ is the positive t value that separates an area of $\alpha/2$ in the right tail of the Student t distribution.



Critical value with right-tail area $\alpha/2$

5 Round-Off Rule

Rounding Confidence Interval Limits

- If using an original set of data values, round the confidence interval limits to one more decimal place than the original data.
- If using summary statistics n, $\bar{x}$, and s, round the confidence

interval limits to the same number of decimal places as the sample mean.

6 Student t Distribution

The Student t distribution is used when the population standard deviation σ is not known.

Key Points about the Student t Distribution

- It has the same general symmetric bell shape as the standard normal distribution.
- It is centered at 0.
- It has more spread than the standard normal distribution, especially for small samples.
- The shape depends on the degrees of freedom.
- As the sample size n gets larger, the t distribution gets closer to the standard normal distribution.

Degrees of Freedom

For the confidence interval methods in this section,

$$\text{df} = n - 1.$$

7 Procedure for Constructing a Confidence Interval for μ

Steps

1. Verify the requirements: simple random sample and normal population or $n > 30$.
2. Since σ is not known, use the Student t distribution.
3. Find the degrees of freedom:
$$\text{df} = n - 1.$$
4. Find the critical value $t_{\alpha/2}$ using technology or Table A-3.
5. Find the margin of error:
$$E = t_{\alpha/2}\frac{s}{\sqrt{n}}.$$
6. Construct the confidence interval:
$$\bar{x} - E < \mu < \bar{x} + E.$$
7. Round the confidence interval limits correctly.

Example 1: Find the critical value $t_{\alpha/2}$ corresponding to a 95% confidence level, given that the sample has size $n = 6$.

Solution

Because
$$n = 6,$$
the number of degrees of freedom is
$$\text{df} = n - 1 = 6 - 1 = 5.$$

For a 95% confidence level,

$$\alpha = 1 - 0.95 = 0.05.$$

So the area in each tail is

$$\frac{\alpha}{2} = \frac{0.05}{2} = 0.025.$$

Using technology or Table A-3 with

$$\text{df} = 5$$

and right-tail area 0.025, we get

$$t_{\alpha/2} = 2.571.$$

Example 2: Listed below are weights, in grams, of randomly selected Reese's Peanut Butter Cups Miniatures.

$$8.639, \quad 8.689, \quad 8.548, \quad 8.980, \quad 8.936, \quad 9.042$$

A package label states that a package of 38 cups has total weight 340.2 g. Therefore, the expected mean weight of one cup is

$$\frac{340.2}{38} = 8.953 \text{ g}.$$

(a) Find the point estimate of the mean weight.

(b) Construct a 95% confidence interval estimate of the mean weight.

(c) Does the package appear to be filled in an acceptable way?

Solution

Part (a): Point estimate

The best point estimate of the population mean μ is the sample mean:

$$\bar{x} = 8.8057 \text{ g}.$$

Part (b): Confidence interval
The sample size is

$$n = 6,$$

so

$$\text{df} = n - 1 = 5.$$

For a 95% confidence level and $\text{df} = 5$,

$$t_{\alpha/2} = 2.571.$$

From the sample data,

$$s = 0.2054 \text{ g}.$$

Now find the margin of error:

$$E = t_{\alpha/2}\frac{s}{\sqrt{n}} = 2.571\left(\frac{0.2054}{\sqrt{6}}\right).$$

$$E \approx 0.2156.$$

Now construct the confidence interval:

$$\bar{x} - E < \mu < \bar{x} + E.$$

$$8.8057 - 0.2156 < \mu < 8.8057 + 0.2156.$$

$$8.5901 < \mu < 9.0213.$$

So the 95% confidence interval is

$$8.5901 \text{ g} < \mu < 9.0213 \text{ g}.$$

Part (c): Interpretation

The desired mean weight is

$$8.953 \text{ g}.$$

This value is inside the confidence interval

$$8.5901 < \mu < 9.0213.$$

Therefore, the package appears to have been filled in an acceptable way.

8 Interpreting a Confidence Interval

Correct Interpretation

A 95% confidence interval means that we are 95% confident that the interval contains the true population mean μ.

Repeated-Sampling Interpretation

If we were to collect many different random samples and construct a confidence interval from each sample, about 95% of those intervals would contain the true population mean μ.

Common Mistake

Do not say: ``There is a 95% probability that μ is in this specific interval.''
After the interval is computed, the true value of μ is fixed. The method has the 95% long-run success rate.

9 Finding the Point Estimate and Margin of Error from a Confidence Interval

Sometimes we are given a confidence interval and asked to recover the point estimate and margin of error.

Formulas

If a confidence interval has lower limit L and upper limit U, then

$$\text{Point estimate} = \frac{L+U}{2}$$

and

$$E = \frac{U-L}{2}.$$

Example 3: A confidence interval for a population mean is

$$12.4 < \mu < 18.6.$$

Find the point estimate and the margin of error.

Solution

The point estimate is the midpoint of the interval:

$$\frac{12.4+18.6}{2} = \frac{31.0}{2} = 15.5.$$

The margin of error is half the width of the interval:

$$E = \frac{18.6-12.4}{2} = \frac{6.2}{2} = 3.1.$$

Therefore,

$$\text{point estimate} = 15.5$$

and

$$E = 3.1.$$

10 What If σ Is Known?

If the population standard deviation σ is known, then we use the standard normal distribution instead of the Student t distribution.

Margin of Error When σ Is Known

$$E = z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

In practice, σ is rarely known. Most of the time, we use s and the Student t distribution.

11 Choosing the Correct Distribution

Which Distribution Should We Use?

- If σ is not known and the population is normal or $n > 30$, use the Student t distribution:

$$E = t_{\alpha/2}\frac{s}{\sqrt{n}}.$$

- If σ is known and the population is normal or $n > 30$, use the normal z distribution:

$$E = z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

- If the population is not normal and $n \leq 30$, use bootstrapping or a nonparametric method.

12 Sample Size Needed to Estimate a Population Mean

Sometimes we want to know how many sample values are needed before collecting the data.

Sample Size Formula

To estimate a population mean μ with margin of error E, use

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2.$$

If the result is not a whole number, always round up to the next whole number.

Important Requirement

The sample must be a simple random sample.

13 What If σ Is Unknown When Finding Sample Size?

When finding the required sample size, the formula uses σ. If σ is unknown, we can estimate it.

Ways to Estimate σ

- Range rule of thumb:

$$\sigma \approx \frac{\text{range}}{4}.$$

- Start and improve: Start collecting data, compute s from the first several values, and use s as an estimate of σ.

- Use prior results: Use an estimate of σ from an earlier study.

Conservative Rule

When estimating σ, it is safer to use a value that is too large rather than too small. A larger σ gives a larger sample size, which is conservative.

Example 4: Assume that we want to estimate the mean IQ score for the population of statistics students. How many statistics students must be randomly selected for IQ tests if we want 95% confidence that the sample mean is within 3 IQ points of the population mean? Assume that

$$\sigma = 15.$$

Solution

For a 95% confidence level,

$$z_{\alpha/2} = 1.96.$$

The desired margin of error is

$$E = 3.$$

We are given

$$\sigma = 15.$$

Use the sample size formula:

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2.$$

Substitute:

$$n = \left(\frac{1.96(15)}{3}\right)^2.$$

$$n = (9.8)^2 = 96.04.$$

Since the sample size must be a whole number, and we always round up,

$$n = 97.$$

Therefore, at least

$$97$$

statistics students must be randomly selected.

Summary

Key Ideas

- The sample mean $\bar{x}$ is the best point estimate of the population mean μ.
- When σ is not known, use the Student t distribution.
- For estimating μ with σ not known,

$$E = t_{\alpha/2}\frac{s}{\sqrt{n}}.$$

- The confidence interval is

$$\bar{x} - E < \mu < \bar{x} + E.$$

- The degrees of freedom are

$$\text{df} = n - 1.$$

- If σ is known, use the normal z distribution.
- To find the sample size needed to estimate μ,

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2.$$

- Always round a required sample size up.